

Module 2: What Is Data? Big Data and the Data Science Workflow

DSC 210 Foundations of Data Science

References: - [Hands-on Introduction to Data Science with Python](#) (CC BY-NC-SA 4.0) - Palmer Penguins (Horst, Hill, and Gorman, 2020; CC0)

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Key Concepts

- Define data, and distinguish **structured** from **unstructured** data using the vocabulary of **cases**, **features**, and **metadata**
- Classify a feature by **measurement scale** (nominal, ordinal, interval, ratio) and determine which operations it permits
- Explain what makes data **big**, and why that mattered for the emergence of data science
- Judge whether a dataset is the *right* data for a question, and explain why acquisition is central
- Trace the **data to information to knowledge** progression and locate each step on the course loop

In this module, it will be helpful to have a dataset handy to make the abstract vocabulary concrete:

Palmer Penguins: Data were collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network.

The dataset contains body measurement data for 344 penguins. There are 3 different species of penguins in this dataset, collected from 3 islands in the Palmer Archipelago, Antarctica. The dataset is included in the **seaborn** Python package, so there is nothing to download.

Class Example 2.1 - Loading the Penguins Data

```
# RUN-TOGETHER
import seaborn as sns
import pandas as pd

penguins = sns.load_dataset('penguins')
# sns.load_dataset() reads the penguins data into a DataFrame:
# the table-shaped structure provided by the pandas library.
# A DataFrame has attributes (such as .shape) and methods (such as .head()).
penguins.head()
```

A working definition: **data is information recorded through observation or measurement.** That is deliberately broad, because data has taken wildly different physical forms across history:

- tally marks carved on bone,
- London parish death records,
- readings streamed from a wearable fitness device,
- every post on a social media platform.

(Aside: whether “data” is singular or plural is genuinely contested. We will not settle it today.)

Activity 2.1 - Data sources (Think-Pair-Share)

Name one source of data that did **not** exist 30 years ago, and one that has existed for centuries. For each, what is being *observed or measured*?

Write both down and keep them. We return to these two sources in Section 4, once we have the vocabulary to say what makes data “big.”

My two sources:

1. New: _____ What it measures: _____
2. Old: _____ What it measures: _____

There are two general classifications of data:

- Structured data: organized in a defined format
 - Typically arranged in rows and columns (sometimes called rectangular data)
 - Examples: a spreadsheet, .csv files, databases, the penguin dataset
- Unstructured data: no specific form or model
 - Examples: Text documents, images, video, audio, etc.

Within a structured (tabular) dataset:

- Features: distinct properties measured
 - Also called variables or attributes
 - Generally the ‘columns’ of a tabular dataset
 - In UWL student records, these could be ‘Academic Year’, ‘Major’, or ‘GPA’
- Cases: unit of observation of a dataset
 - Sometimes called ‘units’ or ‘subjects’
 - Generally the ‘rows’ of a tabular dataset

- In UWL student records, these would be the individual students

Note: For an image, the cases could be the pixel (x, y) location or a pixel numbering and the features could be the RGB color code, or object/no object.

- Metadata: information about the data
 - context, quality, condition, origin, and other characteristics.
 - If data is a book, metadata would contain information about content, author, and publication details.

*This photograph is data. Its metadata is everything the file records **about** the photograph: when it was taken, on what device, at what location, at what resolution, and by whom. Notice that you can often learn a great deal from metadata alone, without ever looking at the image.*

Activity 2.2 - Cases and features of penguins

Recall that we loaded into memory a dataset containing body measurement data recorded for 344 penguins.

In the penguins dataset, the rows are individual penguins and the columns are things measured about them. **Before running the next cell**, predict: how many **cases** (rows) and how many **features** (columns) are there?

My prediction: _____ cases, _____ features.

```
# RUN-TOGETHER
n_cases, n_features = penguins.shape
print('cases (rows)    : ', n_cases)
print('features (cols): ', n_features)

print('the feature names:')
print(list(penguins.columns))
```

Class Example 2.2 - Selecting a single case

Now let's look at a **single case**: one penguin, and all of its feature values.

```
# FILL-IN-LIVE
# Goal: pull out the very first case (one penguin) and see all its feature values.
# together: which position selects the FIRST row? Change the number below.
row_position = 0

first_case = penguins.iloc[row_position]
print(first_case)
```

Activity 2.3 - What got left out?

The penguins table is *structured*. Discuss with your group:

1. What would the **unstructured** version of penguin data look like? Think: what did a field researcher actually record before it became this tidy table?
2. Who decided which measurements became columns, and what got left out?
3. Row 3 of the table is nearly all blank. What are two different reasons a field researcher might return with a mostly empty row?

Determining the type of features we record is the first step in deciding how we visualize the features and directs which analysis tool or model that we use.

Scale Type	Characteristics	Data Type	Operations	Examples
Nominal	Categories without order	Categorical	Classification, Mode	penguin species, island, major
Ordinal	Ordered categories, unequal intervals	Categorical	Sorting, Median, Percentiles	Ratings (poor, fair, good)
Interval	Equal intervals, no true zero	Numerical	Addition, Subtraction, Mean, Standard Deviation	Temperature (Celsius, Fahrenheit)
Ratio	Equal intervals, true zero, meaningful ratios	Numerical	All arithmetic operations	Height, Weight, Age, Income

Activity 2.4 - Does a mean make sense?

Predict, then run. Consider two penguin features: `species` (Adelie, Chinstrap, Gentoo) and `body_mass_g`. For which one does asking for a **mean** make sense? Predict, then we test both.

My prediction:

Class Example 2.3 - Meaningful and meaningless summaries

```
# RUN-TOGETHER
# Ratio feature: a mean is meaningful.
print('mean body mass:', round(penguins['body_mass_g'].mean(), 1), 'g')
```

```
# RUN-TOGETHER
# Nominal feature: a mean is not meaningful.
print('species:', round(penguins['species'].mean(), 1))
```

Python stores features as different data types (called dtype). The *string* data type does not have the attribute mean. What can we do instead with a categorical variable?

```
# RUN-TOGETHER
# Nominal feature: what is the RIGHT summary? Not a mean, but a count.
print(penguins['species'].value_counts())
```

Activity 2.5 - What type is it?

Classify each feature below by measurement scale (nominal, ordinal, interval, ratio). **Three of these are contested.** Be ready to defend your answer.

#	Feature
1	Penguin species (Adelie, Chinstrap, Gentoo)
2	Flipper length in mm
3	The island a penguin was found on
4	A student's class standing (first-year, sophomore, junior, senior)
5	A UWL student ID number
6	Course grade on the A, AB, B, BC, C scale
7	The year a measurement was taken
8	Temperature in degrees Celsius

Part A. For any two features you classified differently, name an operation that is meaningful for one and not the other.

Part B. Run the cell below, then discuss: does knowing how Python *stores* a feature tell you its measurement scale? Where does that reasoning break down?

```
# RUN-TOGETHER
# How Python STORES a feature is not the same as what the feature MEANS.
print(penguins.dtypes)
```

Part C. `body_mass_g` is a numerical feature for which it makes sense to find the average body mass of a penguin.

Warning: Not all features that are recorded as numbers are numerical features.

Suppose we collect the zipcode of UWL students. Zipcodes are stored as numbers. Why are they **not** numerical features?

Part D. State, in one sentence, the question we can ask about any feature to determine its type.

Notes:

Definition: *Big data* are data that are too large, too complex, or too fast-changing to handle manually or with traditional methods. The penguins table (344 rows) is **not** big; we could inspect it by hand. Bigness is usually described by a set of V's:

- **Volume:** sheer quantity (terabytes, petabytes).
- **Velocity:** how fast it arrives (a sensor stream, a live feed).
- **Variety:** many forms at once (text + images + numbers).

Uses of Big Data

Activity 2.6 - Data sources, revisited (Think-Pair-Share)

Return to the two data sources you named in **Activity 2.1**. (If you no longer have them, name two new ones now: one that did not exist 30 years ago, one that has existed for centuries.)

Part A. Rate each source low / medium / high on **Volume** and on **Velocity**.

Source	Volume	Velocity	Big?
New:			
Old:			

Part B. Does either one qualify as “big”? What specifically makes traditional, by-hand analysis break down for it?

Part C. Take the *old* source and describe one change that would make it big. Which V does your change affect?

Note: The *reason* data science emerged as a field is that bigness broke the old tools. When you cannot eyeball the data or fit it in a spreadsheet, you need code, statistics, and algorithms working together. That is the whole motivation for learning the data science workflow.

But that raises the very first practical question of any project, and the subject of the rest of the Module: *where does all this data come from, and can we trust it?*

In most classroom examples, including ours today, a dataset appears by magic: `sns.load_dataset('penguins')` and there it is, tidy and documented. Real projects almost never start that way.

Huber illustrates the point with a figure suggesting that a data science project is often **50 to 99% about getting, cleaning, and processing the data**.

A project usually begins not with a model, and not even with a table, but with a **question**, and then the hunt for data that can actually answer it.

Definition: *Data acquisition* is the process of obtaining or creating data suitable for answering a specific question. Note the last three words: the goal is not *some* data, but data that is **relevant, trustworthy, and suitable** for the problem at hand.

Common sources, each with a catch:

Source	Example	The catch
Public datasets	Kaggle, UCI, census, weather	“Public” does not mean unbiased, current, or fit for <i>your</i> question
Web / APIs	a service’s official API	APIs are stable; scraping is fragile and sometimes disallowed
Internal org data	school records, transaction logs	Privacy, legal limits, access restrictions
Scientific data	data published with a paper	May need real domain knowledge to interpret
Bought data	a commercial dataset	Paying for it does not make it good
Create your own	survey, sensor, experiment	You now own its design, quality, and ethics

Activity 2.7 - Match the source, find the catch

Your group is asked: **do UWL students who use the library more often earn higher grades?**

For each source below, decide (a) which row of the table above it belongs to, (b) what its catch is for *this* question, and (c) whether you would use it.

#	Available source
1	Library door-counter totals, by hour, for the past three years
2	A voluntary survey emailed to all students asking how often they study in the library
3	Registrar records of every student’s GPA

#	Available source
4	Wireless network logs showing which access point each device connected to

Then: none of these four is sufficient on its own. What would you actually need, and what would it take to get it? Does anything about the source concern you?

Key idea: garbage in, garbage out. A dataset can be huge and still be the *wrong* data.

A. Customer Churn: A subscription company wants to answer the question: why do customers cancel their subscription?

You are hired as a data scientist to answer this question. “Don’t worry, we have tons of data.” The company gives you access to millions of rows of website clicks and purchase logs.

B. Wildlife Camera: A motion camera in a forest records day and night for a month to answer the question: *when and where is a particular animal active?*

You are hired as a paid data scientist intern to answer this question. The company gives you hours and hours and hours (a whole month) of video.

Activity 2.8 - Is all data good data? (Think-Pair-Share)

Talk with your group about one or both of the situations above. - How could we provide an answer to the question with the data that is given? - How good of an answer do we give? - Do we have the right data to answer the question? - If not, what data do we need instead?

Note: Nowadays, answers to questions are commonly not limited by having too little data. It is the absence of the **right** data. No algorithm can fix that later.

A dataset is never just “found.”

Even a clean, public dataset was *collected, selected, and structured* by someone who made choices: what to measure, what to leave out, how to categorize. Many datasets have blank recordings of features for some rows. Someone decides which measurements to attempt, and reality decided which ones came back blank.

Data is not neutral

Suppose we have acquired our data. Now what? Here is the heart of the matter, and a mild surprise:

Data science is not really about data.

Data is the raw material. The actual goal is *understanding*, or more formally, **knowledge**. We dig through data because we hope to *learn* something, not because the data itself is beautiful.

What does it even mean to “know” something? A classic answer: you **know** a claim when it is a **justified true belief**, all three at once:

1. you **believe** it,
2. it is **true**, and
3. you have **good justification** for it.

Why all three?

Believing something false is not knowledge. And being *accidentally* right is not either: “I know the next coin flip is heads,” which then happens, was a lucky guess, not knowledge, because there was no good reason behind it.

This is the whole job of data science: to supply the justification, the good reasons, that turn a belief into knowledge.

That is what separates an analysis from a hunch.

Definition: The *data - information - knowledge progression*:

Stage	What it is	Customer churn (A)	Wildlife camera (B)
Data	raw recordings; may be mostly meaningless	millions of click and purchase logs	a month of footage, mostly empty scenery, wind, shadows
Information	data in context; “data with potential significance for a user”	logs joined to <i>which</i> customers cancelled and when	frames tagged: <i>animal X, this location, this time</i>
Knowledge	robust patterns and explanations you can act on	“customers churn ~2 months after a price increase”	“animal X reliably appears at this clearing at dawn”

Notice the raw data at the top of each column is nearly useless on its own. Value is *added* by context and interpretation as you move down.

Consider the wildlife camera column. Most recorded frames are empty forest. What fraction of a month of footage do you suppose is actually *informative* for the question “when is animal X active?” Whatever your guess, it says something about how much work remains after you “have the data.”

Data - Information - Knowledge - Wisdom

Some versions of this progression add a fourth tier, **wisdom**: knowing which questions are worth asking in the first place, and which actions are worth taking once you have the knowledge. We will not treat wisdom formally in this course, but every ethical question from Module 1 lives at that level rather than in the three tiers below it.

A detective starts with **clues** (a footprint, a broken window), each weak on its own: that is *data*.

Connecting and contextualizing the clues yields a coherent picture: that is *information*.

Then the detective weighs competing explanations, discards the unlikely ones, and settles on the account that best fits the evidence: that is *knowledge*.

That last move has a name, **inference to the best explanation**:

1. observe the data,
2. propose hypotheses that could explain it,
3. discard hypotheses that do not fit,
4. compare the survivors,
5. select the most plausible, comprehensive explanation.

We rarely *prove* a theory for certain; we make progress by *ruling out* explanations that fail against the data. Does this sound familiar?

Activity 2.9 - Customer churn (Think-Pair-Share)

Suppose the data shows customers who cancelled were far more likely to have contacted support in their final month.

Part A. Give **two different explanations** that both fit this fact.

Part B. What *additional* data would help you decide between them?

Part C. Suppose you cannot get that additional data. What would you tell the company, and how confident would you be?

Step back and look at the road we just traveled this class:

1. We started with a **question** (when is animal X active? why do customers churn?).
2. We **acquired** data suitable for it, and worried about whether it was the *right* data.
3. We turned raw data into **information** by adding context.
4. We reasoned from information toward **knowledge**, an explanation we have good reasons to trust.

That sequence was not improvised. It is *the data science workflow*.

A blueprint that can guide a project from inception to completion

Examples include: - Circular: CRISP-DM: Cross-Industry Standard Process for Data Mining - Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment - Linear: OSEMN - Obtain, Scrub, Explore, Model, and iNterpret - Tree: - after a certain step, the process might branch out into multiple paths, reflecting the multifaceted nature of some data science projects

The Huber text follows a slightly more tailored version: **ASEMIC** - **A**cquire data - **S**crub data - **E**xplore data - **M**odel data - **I**nterpret results - **C**ommunicate results

9.1 The workflow model for this class

```
ASK  ->  GET  ->  EXPLORE  ->  MODEL  ->  COMMUNICATE
^                                     |
|_____ findings raise new questions _____|
```

- **ASK**: frame the question.
- **GET**: acquire and inspect the data.
- **EXPLORE**: clean and contextualize, turning data into information.
- **MODEL**: find and test patterns (for example, the churn-after-price-increase pattern or the dawn-activity pattern).
- **COMMUNICATE**: interpret and share the knowledge so someone can act on it.

And the loop: acting on knowledge raises the next question, so we go around again.

Real data science is cyclical, not a straight line.

9.2 Reconciling with Huber's ASEMIC

The textbook uses a six-phase version of the same journey, **ASEMIC**. Same road, finer grain.

Our loop	Huber's ASEMIC	In plain words
ASK	<i>(overarching question)</i>	what are we trying to find out?
GET	A cquire	obtain and inspect raw data
EXPLORE	S crub + E xplore	clean it; turn data into information
MODEL	M odel	find/test patterns; build explanations
COMMUNICATE	I nterpret + C ommunicate	decide what it means; share it

Other standard workflows you will hear about (CRISP-DM, OSEMN) are variations on the same theme. Do not memorize the acronyms; remember the *shape*: question, data, understanding, and a loop back.

Every module from here to the final adds rows to one running table. By the last week you will have the whole collection, and the review will be about **choosing** from it: given the feature types you have and the question you were asked, which tool is right?

The columns never change:

Tool	Question it answers	Feature types it needs	Label needed?	Key assumption	How it fails
Mean	What is a typical value?	interval or ratio	no	the values are quantities, not labels	meaningless on nominal data; dragged by skew and outliers
Median	What is a typical value?	ordinal, interval, or ratio	no	the values can be put in order	ignores magnitude, so it can hide extremes
Mode / value_counts	Which category is most common?	any, and the only choice for nominal	no	the categories are discrete	uninformative on continuous data, where every value is unique

Suggested Exercises

1. A regional hospital system keeps the following three collections of material:
 - A table with one row per emergency room visit, recording a patient ID, arrival time, reported symptom, and length of stay.
 - Roughly 90,000 free-text notes typed by attending physicians.
 - A document describing which hospitals were included in the system, the date the records began, and a note that one hospital's intake software changed in March.
 - a. Classify each as structured data, unstructured data, or metadata.
 - b. For the structured collection, identify the cases and list the features. Be careful: the cases are not patients.

- c. Explain why the March software change matters to anyone analyzing the first collection.
2. Classify each feature by measurement scale, and give one summary that *is* meaningful for it.
 - a. The number printed on a hospital room door
 - b. A patient's systolic blood pressure
 - c. Triage priority (immediate, urgent, standard, non-urgent)
 - d. The calendar year a patient was born
 - e. A patient's age in years
 - f. Pain reported on a 0 to 10 scale
 - g. Two of these are stored as numbers but are *not* ratio features. Which two, and what goes wrong if you average them? For (f), explain why reasonable people disagree about its scale.
3. You are asked to evaluate the state of digitization in the schools of your region.

Objective: Assess the current level of digital infrastructure and its utilization in schools.

 - a. Write one specific, answerable question that could stand in for this objective. A good check: could two people disagree about the answer, and could data settle it?
 - b. Name two data sources you would pursue, one that already exists and one you would create. For each, name its row in the acquisition table and state its catch.
 - c. The total dollars each district spent on educational technology over the past five years is public, easy to obtain, and accurate. Explain why it might still be the *wrong* data for your question. Note that “wrong” here does not mean inaccurate.
 - d. Fill in one row of the data, information, knowledge progression for this project. Then state one new question that acting on your finding would raise.
4. A grocery chain installs sensors that record how long each shopper stands in front of each shelf.
 - a. After one month the chain has 4 billion sensor readings. Using the V's, is this big data? Address at least two V's.
 - b. The raw readings are, on their own, nearly meaningless. Describe what would have to be added to turn them into *information*, and then what would have to be established to turn that into *knowledge*.

- c. Give two different explanations for the finding that shoppers linger longest in front of the cereal shelf. What additional data would help you decide between them?
- d. Which step of the ASK, GET, EXPLORE, MODEL, COMMUNICATE loop does part (c) belong to? Name one 5-C criterion from Module 1 that this project should worry about.